

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-02

BIOLOGY

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Topics To Covered

- 1 Basic Features of Root
- 2 Main Functions Of Root
- 3 Modifications Of Root → NOT in JKBOSE.
→ NEET ✓✓

BASIC FEATURES OF ROOT

- ① Root arises from Radicle.
- ② Root does not possess nodes and internodes.
- ③ Root lacks Chlorophyll.
- ④ Roots are +vely geotropic.
- ⑤ Roots are -vely phototropic.

MAIN FUNCTIONS OF ROOT

- ①. Absorption of H_2O and minerals from Soil.
- ②. Roots provide anchorage to plants with Soil.
- ③. Storage of Reserve food material.
- ④. Synthesis of plant growth regulators (PGR'S)

MODIFICATIONS OF ROOT

- ↓
- NOT in Board Syllabus.
 - not in NEET Syllabus.

- ↓
- In some plants, roots change their shape and structure and perform functions other than main role.
 - < Tap root modifications.
 - Adventitious root modifications.

1. Tap Root Modifications

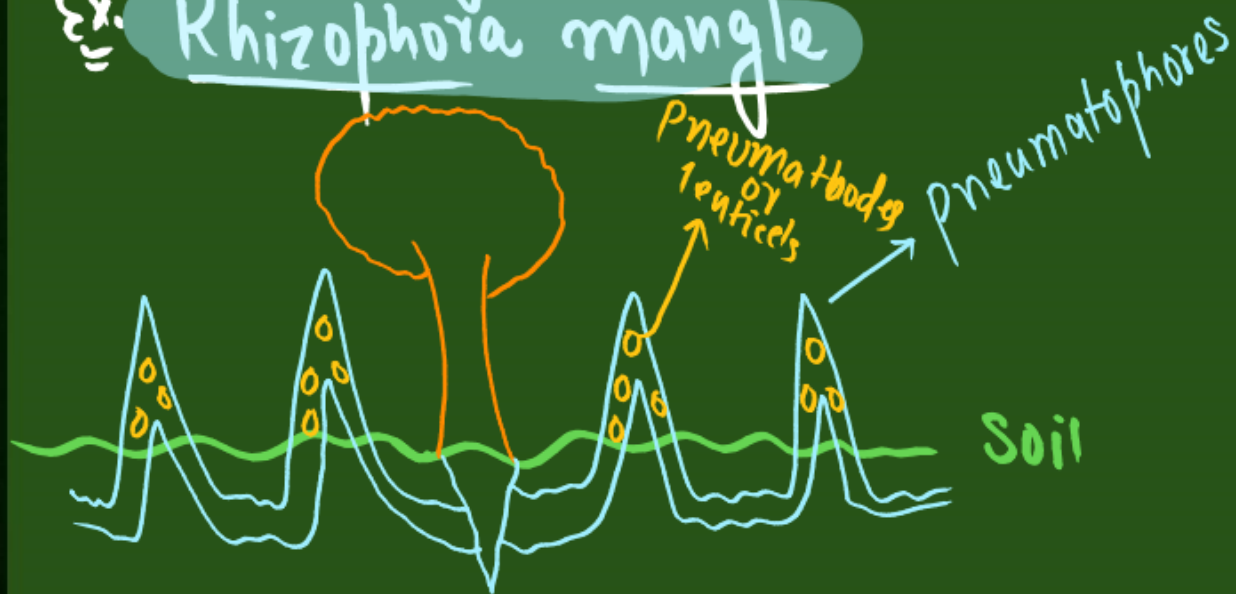
a) For Storage of Food:

Ex. Turnip, Carrot, Raddish
Sugarbeet.

b) For Gaseous Exchange:

→ Seen in halophytes (↑↑ Salt. Conc.)
and mangrooves

Ex. Rhizophora mangle

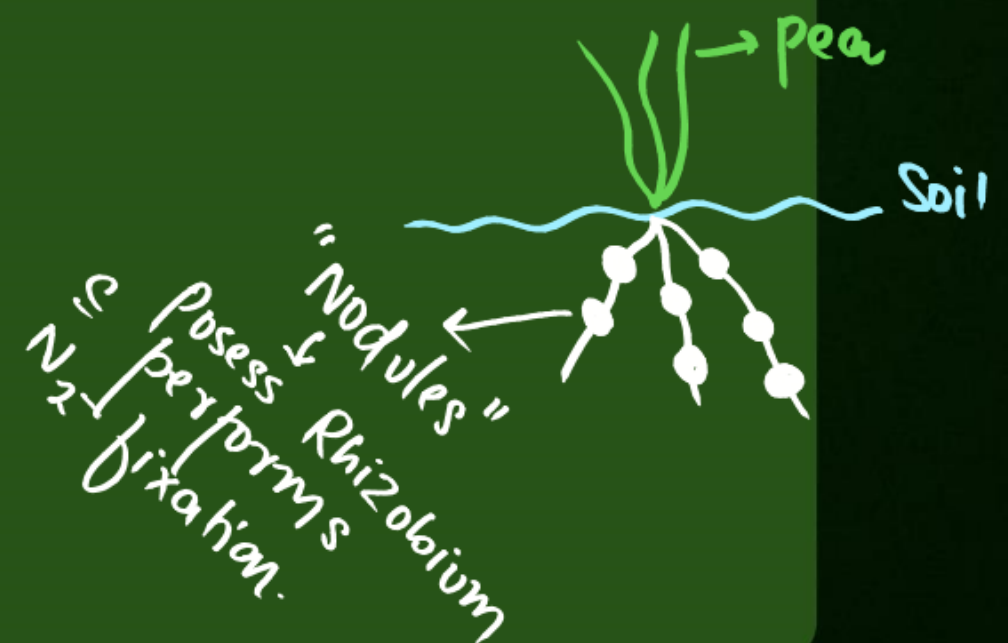


• pneumatophores / respiratory roots — helps in gaseous exchange with the help of lenticels or pneumatophores.

• pneumatophores → - very Geotropic
→ +vely phototropic.

c) For N_2 -fixations:

• In some plants roots form nodules and perform N_2 -fixation Ex. pea.
• These roots are kld as nodulated roots.



2. Adventitious Root Modifications

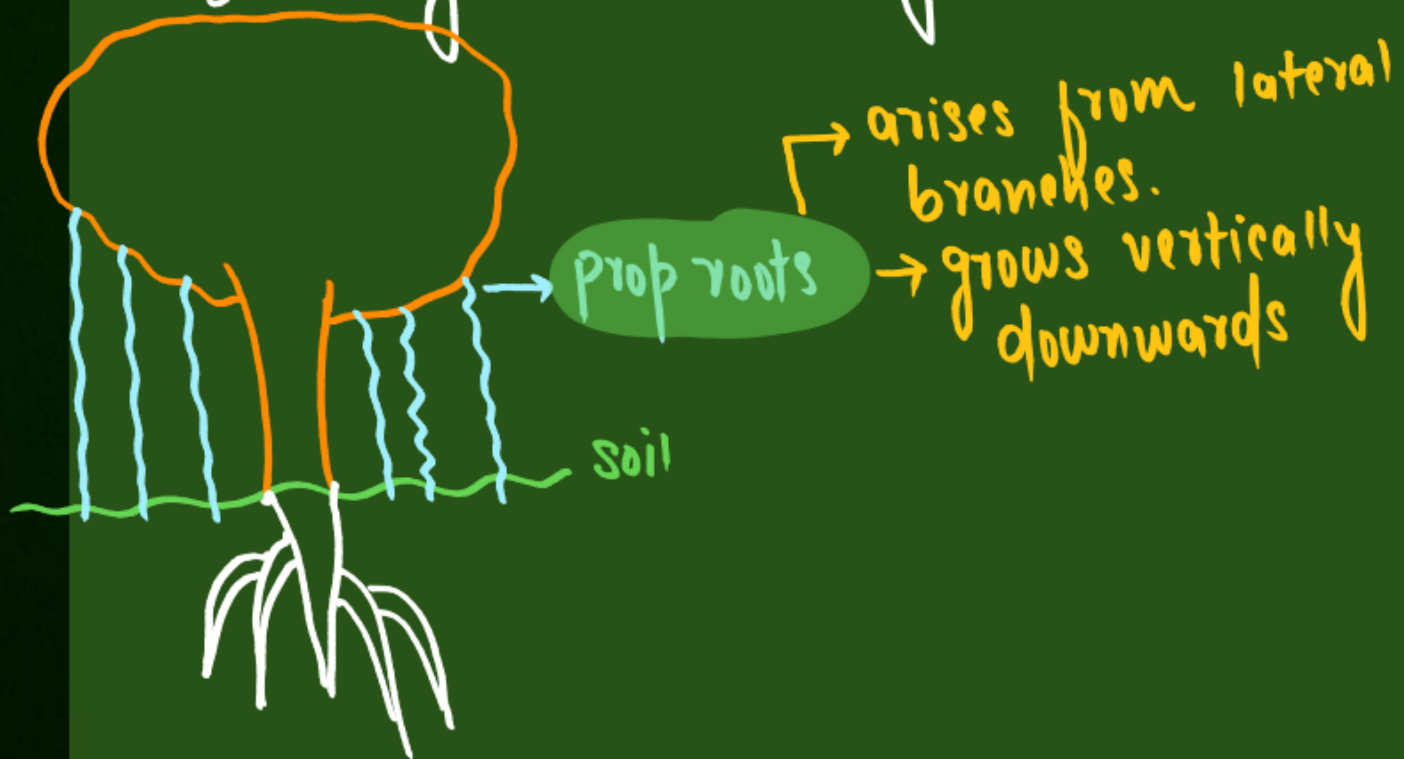
a) For Storage of Food:

Ex. Asparagus, Dahlia, Portulaca, Sweet potato

b) For Mechanical Support:

(i) Prop roots / Pillar Roots:

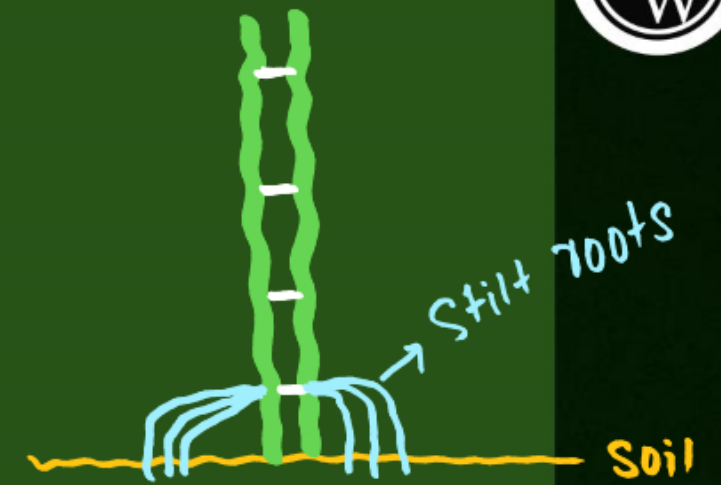
Ex. Banyan (*Ficus bengalensis*)



(ii) Stilt roots / Brace roots :-

Ex. Sugarcane, Maize.

• arise from lower most node of stem.

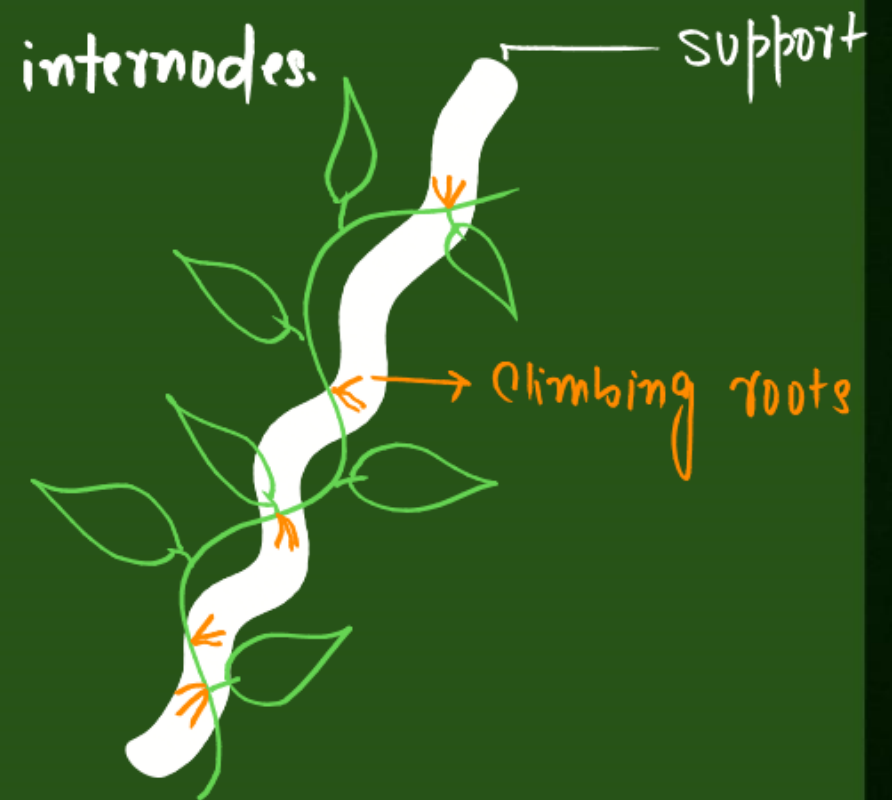


(iii) Climbing roots :-

→ arise from nodes & internodes.

→ helps in climbing.

→ Ex. Black pepper.



Modification of Root

Tap root Modifications

- a) For Food Storage: ex. Raddish, Carrot, Turnip, Sugarbeet.
- b) For Gaseous Exchange: (pneumatophores)
ex. Rhizophora mangle
- c) For N_2 -fixation: (Nodulated roots)
ex. Pea.

Adventitious Root modifications

- a) For Food Storage: Sweet potato, Dahlia, Asparagus, Portulaca.
- b) For Mechanical Support:
 - (i) Prop roots: ex. Ficus bengalensis (Banyan)
 - (ii) Stilt roots: ex. Maize, Sugarcane.
 - (iii) Climbing roots: ex. Black pepper.

QUESTIONS

Pneumatophores are found in:

(2000)

- A** The vegetation which is found in marshy and saline lake
- B** The vegetation which found in saline soil
- C** Xerophytes
- D** Epiphytes

QUESTIONS

Pneumatophores occur in

(2018)

- A** Halophytes
- B** Free-floating hydrophytes
- C** Carnivorous plants
- D** Submerged hydrophytes

QUESTIONS

Assertion: Plants show various kind of root modification. *True*

Reason: Root modification helps in support, food storage and respiration. *True*

- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

QUESTIONS

Which of the following is **not** the main function of the root system ?

- A** Provide anchorage to the plant parts
- B** Synthesis of PGRs
- C** Absorption of water and minerals from soil
- D** Photosynthesis

Thank
You